

The Time of Emergence (ToE) of the Andean Mediterranean sclerophyllous forest of *Quillaja saponaria* (Mol.) and *Lithraea caustica* (mol.) Hox. & Arn

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ABSTRACT

Climate change has intensified in recent decades, leading to increasingly severe droughts. This trend will continue due to the emissions and increasing concentration of greenhouse gas (GHG). One of the main direct climate change consequences is the modification and alteration of the dynamics and function of forest ecosystems, having effects ranging from physiological alterations in plants to definitive changes in their geographical distribution. Mediterranean ecosystems are particularly threatened, where the sclerophyllous forest of central Chile that is considered a biodiversity hotspot, is no exception. Despite that sclerophyllous species present drought adaptation mechanisms, there is no certainty of the climatic minimum thresholds in which it develops, or when climate changes will exceed these thresholds, causing definitive changes in its distribution. To evaluate these changes on the Andean Mediterranean Sclerophyllous Forest vegetation belt of *Quillaja saponaria* and *Lithraea caustica*, the forest was divided into six quadrants with similar climates. In each quadrant, a critical threshold of a simplified water balance (SWB) was defined where the forest has historically developed. The SWB is considered to be a limiting factor in the forest development. Then, the Time of Emergence (ToE) of the forest is estimated, as the time when climate change projections exceed the critical thresholds, by using CMIP6 General Circulation Models (GCM), under SSP2-4.5 (moderate) and SSP5-8.5 (pessimistic) scenarios. Results indicate that the forest would begin to be displaced between 2060 and 2080 under the SSP5-8.5 scenario, particularly in the Talagante, Isla de Maipo, Paine, and Doñihue area, while no distribution changes are presented under SSP2-4.5 scenario.

1. Introduction

Human activities, especially land use change and land degradation, have been the main causes of the loss of global biodiversity (Newbold et al., 2015), affecting forests with high conservation value (Allan et al., 2017; Potapov et al., 2017; Watson et al., 2018). For instance, Mediterranean ecosystems have been strongly impacted in terms of their functioning and composition (Aguayo et al., 2009; Aschmann and Bahare, 1977; Conacher et al., 1998; Schulz et al., 2010). Although climate change acts on a less perceptible time scales, it can significantly affect the structure, composition, dynamics and function of forests over vast areas (Allen et al., 2015, 2010; Clark et al., 2016).

The global climate is changing rapidly, with successive increases in temperature since the pre-industrial era, leading to drought events and heat waves of increasing intensity, severity, and duration (IPCC, 2019). For the coming decades, climate projections indicate that increasing

temperatures and extreme climate events are expected, and several regions of the world expect a reduction in rainfall (Dai, 2013; Huang et al., 2016). This is having multiple impacts on biodiversity and natural ecosystems (IPCC, 2019; Scheffers et al., 2016). In particular, Mediterranean regions are more prone to droughts, and their ecosystems are considered extremely vulnerable to global changes (Schröter et al., 2005), given the observed droughts and projected reductions in rainfall (Polade et al., 2017; Rojas et al., 2019), because they are the ecotone between arid and temperate zones (Barbeta et al., 2015; Huang et al., 2016).

Globally, the Mediterranean climate regions represent less than 5% of the Earth's surface, but host almost 20% of its total biodiversity (Cowling et al., 1996; Lefèvre and Fady, 2016). The Chilean Mediterranean forest is the only Mediterranean-type ecosystem in South America, and like other similar areas it is characterized by a seasonal climate with hot and dry summers, and cold and wet winters, with

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significant inter-annual and spatial variability in rainfall and humidity (Armesto and Arroyo, 2007). Chilean Mediterranean forest is listed as a high priority biodiversity hotspot for conservation (Funk et al., 2010; Myers et al., 2000). It is home of three of the 200 highest global priority ecoregions for conservation (Olson and Dinerstein, 2002), and contains more than 2,400 different plant species, from which 23% are endemic (Cowling et al., 1996). Chilean sclerophyllous forest is inserted in the northern zone of the Chilean Mediterranean region, which is the most populated in the country, with a highly endemic type of vegetation that supports unique biotic communities (Alaniz et al., 2016). The sclerophyllous forest is strongly threatened by a growing agricultural industry, livestock, forest extractions and urban pressure (Schulz et al., 2010), forest fires and climate change (Bowman et al., 2019; González-Reyes et al., 2017).

The sclerophyllous forest is currently under pressure due to Central Chile megadrought (Garreaud et al., 2021), and climate projections show that it should get dryer in the future (Chadwick et al., 2018). One of the main direct consequences of climate change is the modification and alteration of the dynamics and functioning of forest ecosystems (Moore and Allard, 2008; Norby et al., 2007). Although it has been shown that plants are capable to adapt to changes in the climate system through responses at morphoanatomical, physiological, and molecular levels (Ahuja et al., 2010; Reyer et al., 2013), the real concern is that the rate at which the climate is changing during this century may outpace that of evolutionary change and genetic adaptation (Klein et al., 2013). This could force different species to modify their distribution towards latitudes and longitudes different from the current ones, which would lead to changes in the distribution and abundance of species (Parmesan, 2006). However, it is possible that the capabilities of certain ecosystems to migrate or to survive under the new adverse environmental conditions may be exceeded (Gavilán, 2008; Gray, 2005), leading to a decline, decrease or direct extinction of some species and natural populations (Thomas et al., 2004). According to Reyer et al. (2013), Mediterranean sclerophyllous vegetation may be particularly threatened, as evergreen species are considered to be slow adapters to changing environmental conditions. Thus, these species could respond to climate change by migration, if its migration rate allows them this type of adaptation, rather than by evolution or adaptation.

Knowing the affinities, behavior patterns and responses of species to climate conditions can have important practical repercussions, mostly for the land management, forest conservation, and ecosystem restoration (Gutiérrez and Trejo, 2014). For instance, for an effective long term conservation and restoration plan, the design and implementation must consider not only current climate, but also climate change, and other environmental and social factors that may influence it (Aronson et al., 2020; Chazdon and Brancalion, 2019). In some cases, restoration to the original state of an ecosystem, may not be possible due to climate change, requiring rather a “renovation” of the site with target species that reflect the appropriate climatic conditions for both the present and the future (Butterfield et al., 2017; Morecroft et al., 2019), so that they can have a good response to climate change (Wilsey, 2021).

There are several studies related to tree growth under climate change and drought conditions in the northern hemisphere (e.g. Cailleret et al., 2017; Camarero, 2018; Cavin and Jump, 2017; Gazol et al., 2018, 2017a, 2017b; Pretzsch et al., 2013; Serra-Maluquer et al., 2019). However, similar studies are scarce in the extratropical forest ecosystems of the southern hemisphere, and even scarcer for sclerophyllous forest in Chile (Marquet et al., 2019).

Of the studies that have focused on the climate change impacts on vegetation in Chile, including the Mediterranean ecosystems, some have focused on species distribution modelling and/or ecosystems at a given future date (e.g. Arroyo et al., 2005; Bambach et al., 2013; Marquet et al., 2009; Plissock et al., 2012; Santibañez et al., 2014), other have estimated the levels of risk or vulnerability of ecosystems (e.g. Marquet et al., 2009; Miranda et al., 2020a; Muñoz-Sáez et al., 2021; Plissock and Uribe, 2020; Santibañez et al., 2014; Santibañez et al., 2013), among

other studies (Fuentes-Castillo et al., 2019; Matskovsky et al., 2021; Miranda et al., 2020b) revealing mostly the locations and types of vegetation that are most vulnerable to climate change. However, to the best of our knowledge, there are no studies that have focused on finding when significant changes in the distribution of a species or forest ecosystem will occur. To address this type of question, studies often estimate the Time of Emergence (ToE), defined as the time when the climate change signal emerges from the noise of natural variability (Hawkins and Sutton, 2011). The ToE has been studied mainly for variables such as precipitation (Nguyen et al., 2018) and temperature (Mora et al., 2013) or both (Chadwick et al., 2019). Despite this, it is not limited to the use of other variables such as sea levels (Lyu et al., 2014), heat waves (Lopez et al., 2018), aridification (Park et al., 2018), streamflows (Zhuan et al., 2018), or the obsolescence of water reservoirs (Chadwick et al., 2021). Recently, Rojas et al. (2019) assessed the ToE of precipitation for the development of the four main crops worldwide (wheat, soybean, rice and corn), as far as we are aware this is the only study that estimates the ToE for plants development. This shows the need of further studies about the implications of climate change and the ToE over vegetation, and plants in general, and in particular it shows that ToE has not been studied in forest species and ecosystems.

The objective of this study is to respond to this need, by understanding when and where significant changes due to climate change will occur in the Andean Mediterranean Sclerophyllous Forest vegetation belt of Quillaja saponaria and Lithraea caustica described by (Luebert and Plissock, 2017). Note that a vegetation belt, used in Chile, is a location or zone characterized by a set of plant communities with a uniform structure and physiognomy, located under homogeneous meso-climatic conditions, which occupy a determined position along an elevation gradient, at a specific space–time scale. For this, we propose a new easy to use methodological approach that can be conducted by professionals and policy makers in the management and conservation of forest resources field; and in order to adapt the ToE detection for assessing vegetation. The conventional ToE detection method defines thresholds purely under statistical criteria of the studied variable, which not necessarily represents the critical thresholds of climate change impact, hence here we want to define a ToE detection threshold based on the forest development. The method consists of two steps: first, the definition of critical thresholds of a climatic variable that defines the development of the forest in a site; and second, to project the climatic variable considering climate change scenarios to find out when and where the previously defined thresholds are exceeded. The moment when the projected variable exceeds the threshold is defined as the ToE. In this way it is possible to quickly, clearly, and easily assess the information that will be useful for planning and prioritization of mitigation, conservation, restoration and management programs that involve the forest under study. The proposed methodology considers the possibility of using different limiting climatic variable, and thresholds, according to the adaptive characteristics of each forest.

This study is relevant, since the vegetation belts that make up the sclerophyllous forest, presents a great vulnerability because of human presence and the bioclimatic stress expected for this century (Santibañez et al., 2013). The Andean Mediterranean Sclerophyllous Forest of *Q. saponaria* and *L. caustica* is a forest that projected the least bioclimatic stress by 2050, so knowing the ToE of this belt allows an approximation of the ToE of the sclerophyllous forest in general. For the analysis, five future climate projections were used, from the latest generation of General Circulation Models (GCM), under two climate change scenarios (SSP2-4.5 and SSP5-8.5).

2. Methodology

2.1. Study area

The study area corresponds to the Andean Mediterranean Sclerophyllous Forest vegetation belt of *Quillaja saponaria* and *Lithraea*

caustica (Fig. 1), one of the eight vegetation belts that make up the Sclerophyllous Forest formation described by Luebert and Plissock (2017). This covers an area of 4,534 km² and comprise the lower western slopes of the Cordillera de los Andes and the eastern slopes of the Cordillera de la Costa. The area is located between 32°50'S and 34°45'S, 70°10'W and 71°24'W, and between 200 and 1700 m.a.s.l., constituting two surfaces separated from each other by the intermediate depression of central Chile. It is inserted in a Mediterranean-type climate, characterized by winter rainfalls and summer droughts (Armesto and Arroyo, 2007), with average annual rainfall ranging

between 360 and 1100 mm, minimum temperatures ranging from 0 to 6.5 °C and maximum temperatures ranging between 26 and 33 °C (Santibáñez, 2017). The vegetation belt is typically dominated by *Lithraea caustica*, *Quillaja saponaria*, and *Kageneckia oblonga*, while other species such as *Cryptocarya alba* are locally abundant in more humid sectors (Luebert and Plissock, 2017).

Since 2010 Central-Southern area of Chile, which includes the Andean Mediterranean Sclerophyllous Forest vegetation belt of *Quillaja saponaria* and *Lithraea caustica*, has been affected by a megadrought, which is an unprecedented period of drought that has registered a

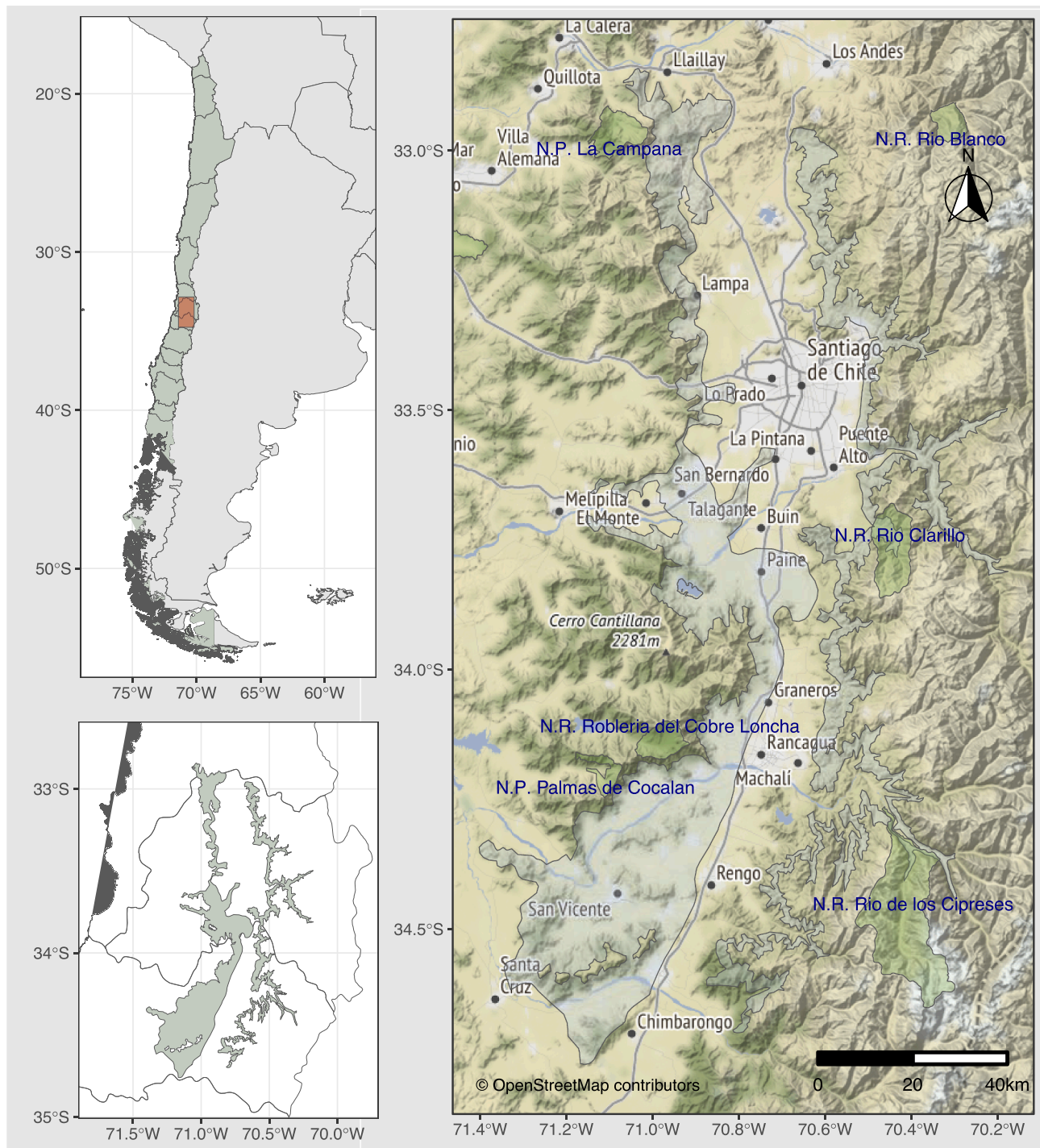


Fig. 1. Map of the Andean Mediterranean Sclerophyllous Forest of *Quillaja saponaria* and *Lithraea caustica* []. adapted from Luebert and Plissock (2017)

rainfall deficit around 30% (Boisier et al., 2016). This megadrought is in part product of the warming of the subtropical Southeastern Pacific, and due to the so-called southern spot, which with the reduction in rainfall in the Central tropical Pacific, and the weak positive trend of the Southern Annular Mode (SAM), are the main drivers of the precipitation reductions in Central Chile (Garreaud et al., 2021, 2020). Although the southern spot and its teleconnections may arise from natural variations, the current warming rate is exceptional (Garreaud et al., 2021), and anthropogenic forcing is estimated to explain more than the 25% of the megadrought (Garreaud et al., 2020). However, climate projections point to an increasing trend of droughts events, that should become more recurrent by the end of the 21st century (Pachauri et al., 2014), which could lead to drought conditions being the average by then (CR2, 2015).

2.2. Meteorological data and general Circulation Models (GCMs)

The meteorological data is the main input data used in this study, which consists in gridded observed, GCM historical, and GCM projected climate (see green blocks in the flowchart of Fig. 2). Gridded monthly observed precipitation (Fig. 3a) and temperature (Fig. 3b) data, was obtained from the CR2MET V2.0 product. CR2MET is the regionalization and statistical corrections of large-scale information provided by the re-analysis ERA-Interim of the European Center for Medium-Range Weather Forecasts (ECMWF) (DGA, 2017). This product contains daily and monthly precipitation, and maximum and minimum temperature, between the years 1979–2019, and has a 0.05° latitude-longitude resolution (~5 km) for continental Chile (Boisier et al., 2018).

The CMIP6 (Coupled Model Intercomparison Project Phase 6) GCMs were used, which are the latest generation of climate models (Eyring et al., 2016). GCMs from two climate scenarios were used, the SSP5-8.5 (extreme) and SSP2-4.5 (moderate). The spatial resolutions of these GCMs is too coarse to be used locally (ranging between 100 and 300 km), hence a downscaling and bias correction is performed (section 2.5).

2.3. Critical thresholds of the forest

Water stress is considered the most limiting factor for plants in Mediterranean climate zones, and is aggravated by other stresses such as high temperatures and solar radiation (Petit et al., 2005). This can be seen reflected in the various adaptive strategies of the species against the recurrent water droughts, which are related to their ability to survive under conditions of water stress (Santibáñez et al., 2013). These were considered to define the critical thresholds where the Andean Mediterranean Sclerophyllous Forest of *Q. saponaria* and *L. caustica* develops, and

a simplified water balance (SWB) was defined as a limiting factor for the future development of the forest, similar to the one considered to study droughts by Vicente-Serrano et al. (2010). The SWB is estimated as the difference between precipitation (P) and potential evapotranspiration (PET), which is a simplified representation of the water surplus or deficit per month (i.e. the precipitation is the gridded precipitation, see orange block in Fig. 2).

Since PET data was not available, it was calculated by using Thornthwaite (1948), also used by Vicente-Serrano et al. (2010). This has the advantage of being easy to estimate, since it only requires monthly mean temperature data and the latitude (note that temperature is the gridded temperature, Fig. 2). The reason for adopting this simple method to calculate PET is that according to (Mavromatis, 2007), similar results are provided by this and other more complex methods. The SWB is estimated between the years 1979 and 2019 for each grid point (Fig. 3c). The monthly SWB is aggregated into an annual scale by a moving sum, leaving the simplified water balance at an annual scale (SWB₁), as presented in Eq. (1).

$$SWB_1 = \sum_{i=1}^{12} P_i - PET_i \quad (1)$$

The responses of Central Chile Mediterranean forest to climate change has not been well studied yet (Matskovsky et al., 2021), hence, there is no information regarding the climatic thresholds where the forest is able to settle. For this reason, we propose using an unfavorable value of the simplified water balance (SWB) as a critical threshold. The critical threshold is defined for each grid point, as a way of considering the different adaptive responses that the species of the forest may present because of the mesoclimatic conditions in which they have developed. A threshold corresponding to the 25th percentile (P25%) of SWB is considered to be critical in terms of the composition, structure, dynamics and/or functions as a result of changes in climate, as documented in other places (Allen et al., 2015, 2010; Clark et al., 2016). Another threshold corresponding to the 10th percentile (P10%) of the SWB was also defined, under the assumption that, with values of SWB under P10%, the development of the forest in some places would no longer be possible. These critical thresholds are represented in the blue box obtained from the percentiles of SWB in the flowchart from Fig. 2.

Unfortunately, there is no certainty regarding the response capacity of the forest according to its origin. A study in *Quillaja saponaria* seedlings indicated that the responses to water stress are rather uniform throughout its distribution (Magni et al., 2019). Considering this possibility, an alternative methodology was developed in which a fixed threshold was defined for the entire forest (i.e. the alternative critical

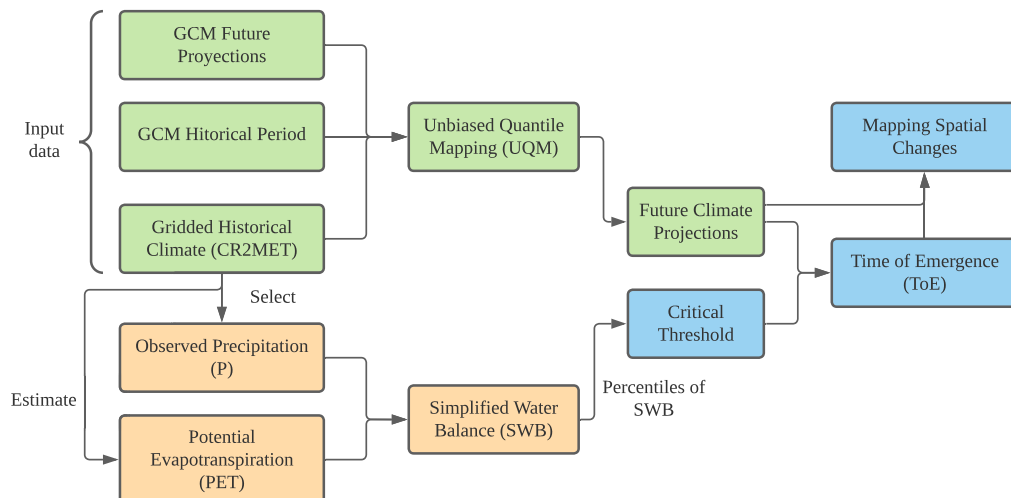


Fig. 2. Flowchart schematics of the methodological steps.

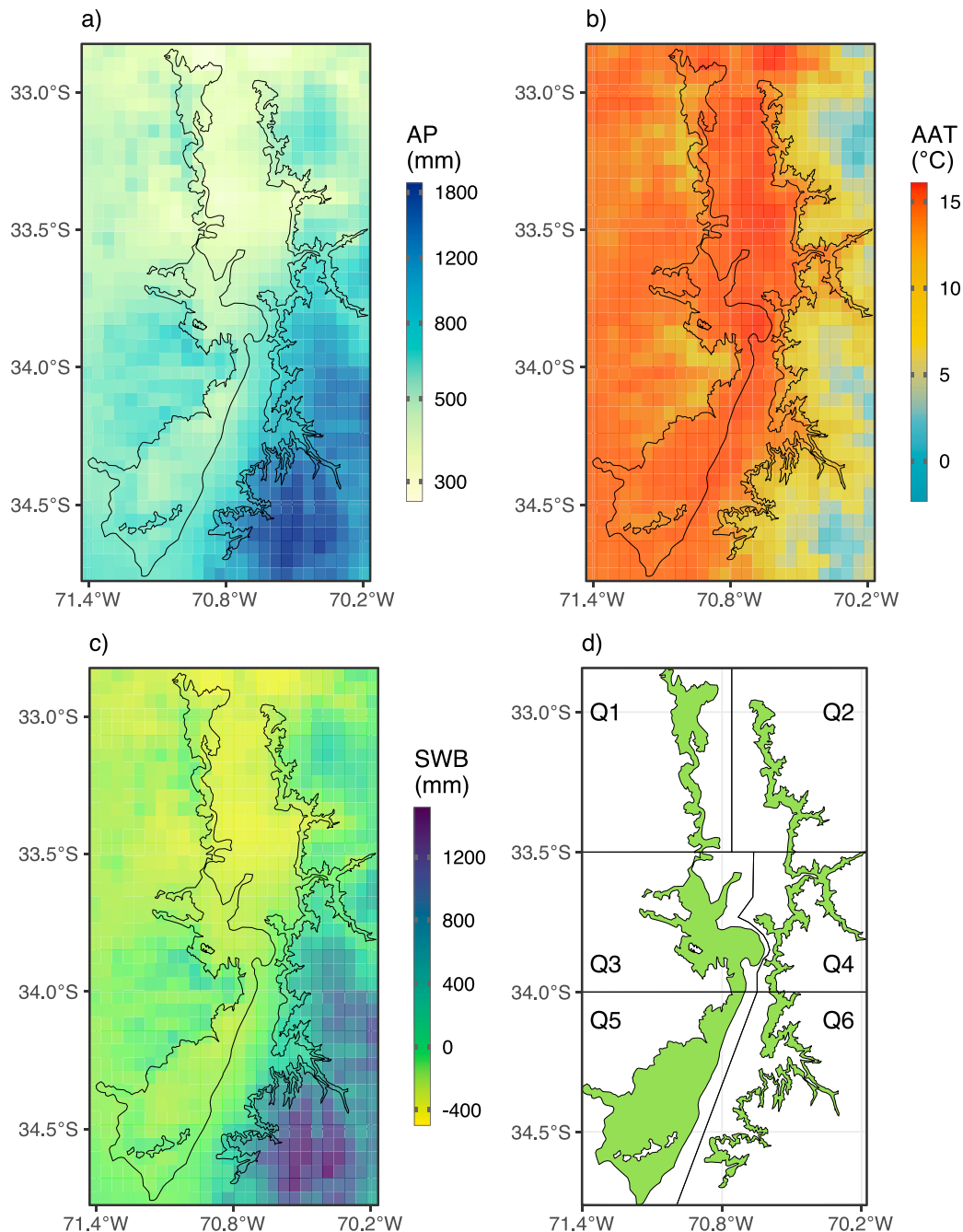


Fig. 3. Division of the study area into quadrants according to the climatic variation of the forest manifested in the historical period (1979–2019). Where a) Annual precipitation (AP), b) Average annual temperature (AAT), c) Simplified water balance (SWB), and d) Division of the forest into quadrants (C1 to C6).

threshold of P50% and P10% gives values of -383 and -473 mm of SWB, for more detail we refer the reader to [Supporting Information Appendix S2](#)). The selected fixed threshold was defined considering the climate of areas where the forest has been strongly affected, with browning symptoms in recent years as a result of the megadrought ([Miranda et al., 2020b](#)). Therefore, the methodology presented in this article represents a general approach for the evaluation of the ToE of a forest due to climate change, seeking to respond and understand in a simple way the response capacity of a forest to climate. In addition, as an alternative method, a more specific approach to the forest under study is presented, based on the scarce information on the browning that the forest has presented due to drought.

The responses of an ecosystem to environmental changes can be

varied depending on factors such as the intensity and frequency of these events ([McDowell et al., 2008](#)), in addition to the ecosystem type, longevity and mechanisms of its species to cope with such changes ([Gutschick and BassiriRad, 2003](#); [Hamrick, 2004](#); [Tuberosa et al., 2003](#)). As climate change advances and the water deficit increases, the effects on the vegetation become evident, causing dysfunctions in the physiological processes, which can lead to the detachment of leaves and buds ([Rood et al., 2000](#)), hydraulic failures ([Anderegg et al., 2013](#)), or reduction of the absorption and accumulation of CO_2 ([Ciais et al., 2005](#)). However, the longevity of the trees can delay the changes in distribution of a forest due to climate change, unless the intensity is such that it causes massive deaths of individuals, as it has already been documented in other regions of the world ([Allen and Breshear, 1998](#); [Berg et al.,](#)

2006; Breshears et al., 2005; Condit et al., 1995; Gitlin et al., 2006; Hanson and Weltzin, 2000; Shaw et al., 2005). These widespread mortality events have the ability to radically transform landscapes on a regional scale in short periods of time (sub-decadal), with significant implications for ecosystem structure and function (McDowell et al., 2008). For this reason, the 10th percentile was determined as the critical threshold, since it represents rare events, with a low probability of occurrence, and defined as extreme according to the IPCC (2007).

2.4. Gcms used

The GCMs of the CMIP6 combine the Shared Socioeconomic Pathways (SSP, O'Neill et al., 2014), which have a narrative of possible mitigation and adaptation scenarios (O'Neill et al., 2017), and the Representative Concentration Pathways (RCP, Moss et al., 2010), which describe a set of alternative trajectories for atmospheric concentrations of greenhouse gases (van Vuuren et al., 2011). Together they form the SSP-RCP scenarios.

For each scenario more than 40 climate projections were obtained from 22 GCMs and their realizations (Table S1). Among these, a subgroup of five GCMs were selected, considering their precipitation and temperature changes, between the future (2071–2100) and historical (1981–2010) periods (Fig. 4 and Table 1).

For the GCM selection, the central point of the study area (33° 46' 30" S, 71° 01' 00" W) is considered. Then five GCMs were visually selected, two from SSP2-4.5 and three from SSP5-8.5, which map the changes in temperature and precipitation in the study area. Fig. 4 shows the change rates of the GCMs for the study area in which the selected ones are marked in green (more detail in Table 1). The selected GCMs appear as input data in Fig. 2.

2.5. Downscaling

GCMs are originally designed to simulate large-scale phenomena (~100 km), but not to be used directly at the local scale studies. To be able to use climate change data at a local scale a scaling process is needed (Wilby and Wigley, 1997). These requires two processes: 1) the downscaling itself and 2) a bias correction. The former requires going from the coarse grid to a finer grid, while the latter is centered in eliminating biases coming from different assumptions, such as GCM parameterization. Downscaling methods are divided into two large groups: dynamical downscaling (Regional Climate Models), which downscale GCMs by simulating local physics, considering GCMs as boundary conditions; and statistical downscaling, which establishes statistical relationships between GCMs and local climate. This study uses

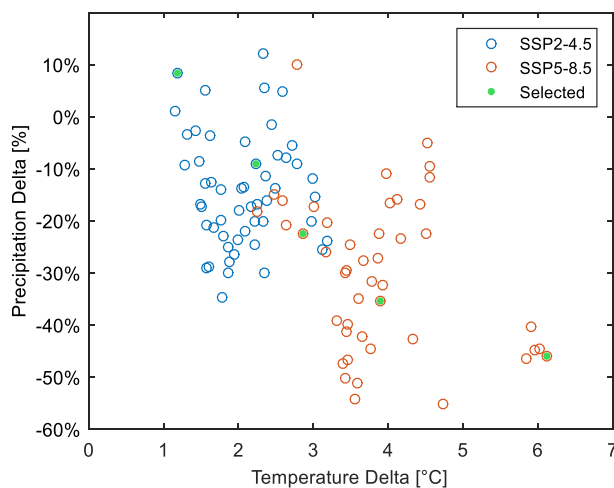


Fig. 4. Precipitation and temperatures GCM changes between a future period (2071–2100) and historical period (1981–2010).

Table 1

Precipitation and temperature changes of the selected GCMs, between a future (2071–2100) and historical (1981–2010) period.

Scenario	SSP2-4.5	SSP2-4.5	SSP5-8.5	SSP5-8.5	SSP5-8.5
GCM	NorESM2-LM	CESM2-WACCM	INM-CM4-8	IPSL-CM6A-LR	CanESM5
Realization	r2i1p1f1	r3i1p1f1	r1i1p1f1	r1i1p1f1	r2i1p1f1
ΔT °C	1,2	2,2	2,9	3,9	6,1
ΔP	8,5%	−9,1%	−22,4%	−35,3%	−46,0%

statistical downscaling because they are computationally less intensive than dynamical ones (Fowler et al., 2007).

For the downscaling method we used bilinear interpolation between the GCM grid and the local climate grid. Regarding the bias correction one of the most widely used is the Quantile Mapping (QM) method (Wood et al., 2002). However, the standard QM method has been shown to have problems preserving the raw GCM changes (Cannon et al., 2015; Chadwick et al., 2018; Maurer and Pierce, 2014). Because of these, a variety of alternative trend preserving quantile mapping methods exist: Detrended Quantile Mapping (DQM, Bürger et al., 2013), Quantile Delta Mapping (QDM, Cannon et al., 2015), Scale Distribution Mapping (SDM, Switanek et al., 2017). In this paper we used the trend preserving quantile mapping called Unbiased Quantile Mapping (UQM), because it has shown good results in the study zone (Chadwick et al., 2023) and is able to preserve raw GCM changes in the mean and standard deviation of precipitation and temperature. The UQM has some similarities with the unbiased mapping proposed and used by Chadwick et al., (2021, 2019, 2018) capable of preserving raw GCM changes. In particular we used the ClimQMBC package which incorporates the UQM as well as other quantile mapping methods (for more detail we refer the reader to Chadwick et al., 2023 and <https://github.com/saedoquilongo/climQMBC>). After the downscaling and UQM bias correction, one obtains local gridded climate projections (Fig. 2).

2.6. Detection of the time of Emergence (ToE)

For the detection of the ToE in sclerophyllous forest we used an approach that identifies the moment in which the projected variable into the future exceeds a previously defined critical threshold. A similar approach was adopted by Chadwick et al. (2021) to estimate the ToE of reservoir performance indexes. In this study the projected variable analyzed is the SWB for the vegetation belt of interest. The future climatic projections are used to estimate the SWB projection for each GCM. Given that initially SWB is at a monthly scale, just as in the observed period, the monthly SWB projections are aggregated into annual scale by a moving sum, leaving the simplified water balance at an annual scale (SWB₁).

The ToE detection of the SWB (blue boxes in flowchart of Fig. 2) was performed per grid point under climate change conditions, in order to identify the spatio-temporal changes of the forest in terms of composition, structure and functions (when intercepting with the critical threshold P25%); and where potentially the development of the forest ceases to be feasible in some places (intercept with the critical threshold P10%). In particular, decision makers are interested in identifying the first areas where the thresholds are exceeded, since those sites are the most susceptible to climate change, and may represent places where forest migration may occur first (i.e., mapping the spatial changes, Fig. 2). For this purpose, ToE maps are built, also to facilitate the results presentation, the forest was divided into six quadrants (Q Fig. 3d).

In particular, the ToE of the sclerophyllous forest is detected when the projected SWB (obtained in section 3.1) overpasses the critical thresholds (presented in section 3.2). To avoid the noise that the SWB might have and detecting an apparent ToE, we used the 30-year moving average of the SWB₁, which from now on will be call SWB₃₀, hence the ToE of the forest will be detected when SWB₃₀ exceeds the thresholds.

Note that a 30-year window represents the climate that the forest is facing and not the variability of SWB.

3. Results

3.1. Climatic projections

The SWB, used to detect the sclerophyllous forest ToE, is formed from the climate change projections. For this, we start by showing precipitation (Fig. 5) and temperature (Fig. 6) projections for the GCM INM-CM4-8, which has an intermediate rate of change (Table 1 and Fig. 4). Fig. 4a and 5a have the time series of the average climate per quadrant, along with their linear trends. Additionally, 20-year windows are used to present the spatial changes in climate in Fig. 5b and 5b. More detail on other climate projections for the study zone is presented in

Fig. S2 to S9 (Supporting Information Appendix S3).

A precipitation reduction is shown in Fig. 5, which is in line with other climate projection studies for Mediterranean areas (Giorgi and Lionello, 2008; Sala et al., 2000). Given that the sclerophyllous forest is found in a Mediterranean climate, we expect that it will have more frequent episodes of drought. Precipitations presents similar trend for quadrants 1 to 3 (Fig. 5), with annual values of 450 mm at the beginning of the historical period, which decrease to ~ 300 mm by 2100. Quadrants 4 and 5 show slightly higher amounts of annual precipitation, beginning the historical period with values close to 600 mm, which decrease to ~ 400 mm by the end of the century, presenting a higher reduction (in mm/year) than Q1 to Q3. Finally, Q6 presents amounts of precipitation considerably higher than the rest, beginning the historical period with ~ 1100 mm, which are reduced to close to 800 mm at the end of the century, which is the highest rate of precipitation reduction

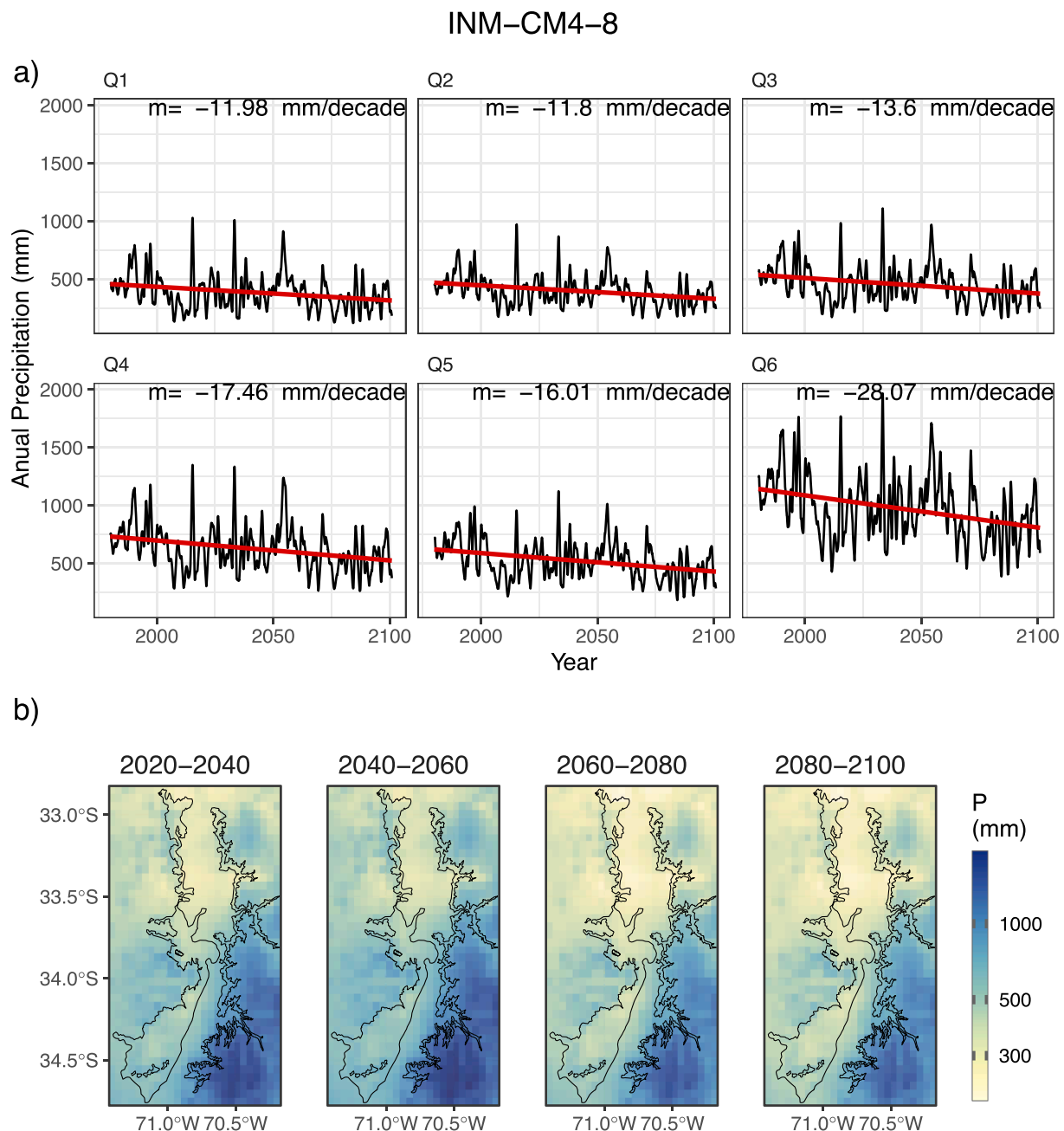


Fig. 5. Annual precipitation of INM-CM4-8. A) Average time series for each quadrant, and its linear trend (red line). B) Spatial changes in precipitation using 20-year windows between 2020 and 2100.

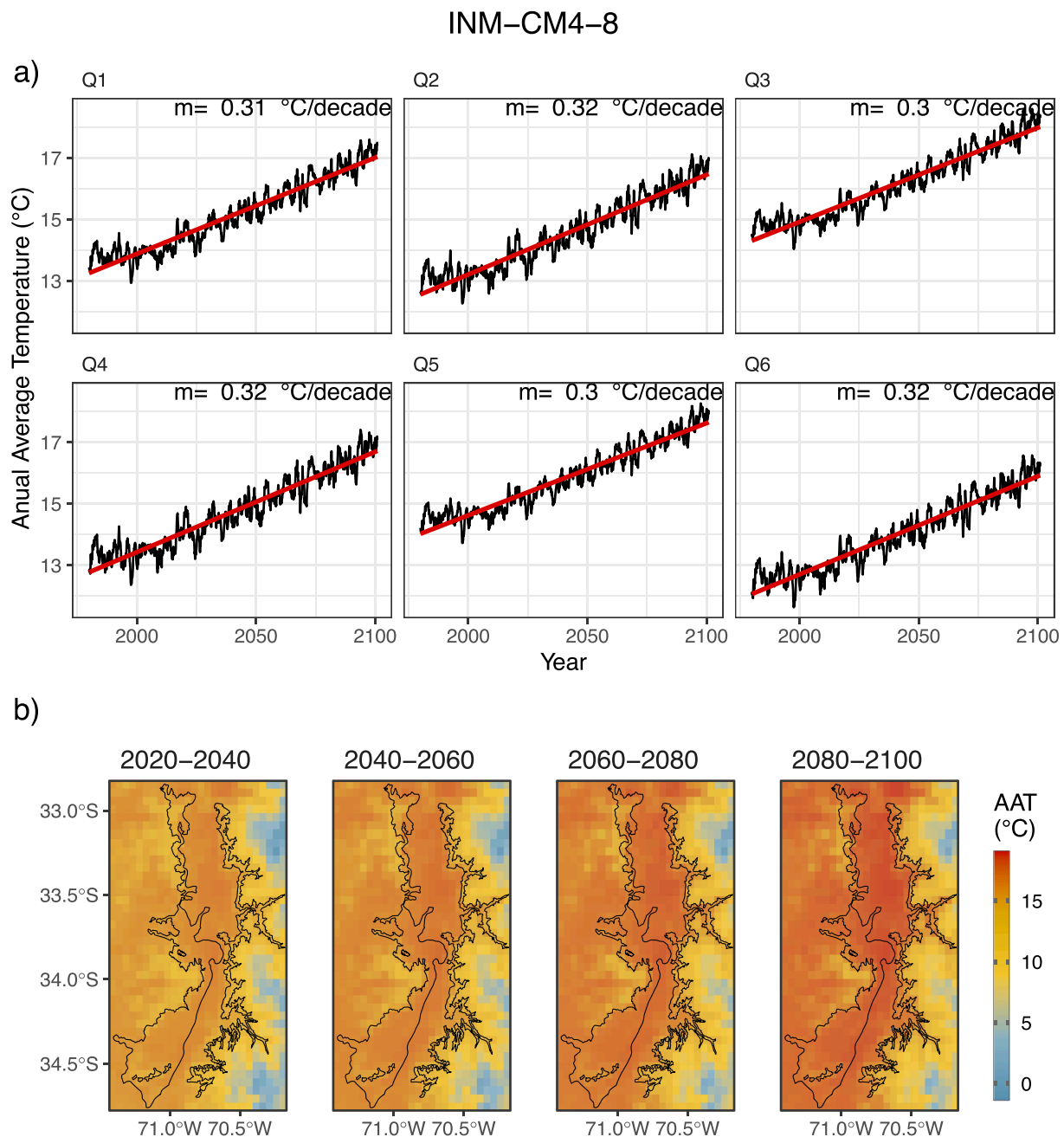


Fig. 6. Average temperature of INM-CM4-8. A) Average time series for each quadrant, and its linear trend (red line). B) Spatial changes in temperature using 20-years windows between 2020 and 2100.

(mm/year) in the forest. Overall, the percentage of change of precipitation is similar between quadrants, hence there is a greater reduction in absolute terms in the rainiest areas (i.e. mm/year).

Fig. 6 shows the temporal (Fig. 6a) and spatial (Fig. 6b) changes in annual temperature. Temperature starts with values between 12 and 15 $^{\circ}\text{C}$ depending on its location, which increases 4 $^{\circ}\text{C}$ by the end of the century. This temperature increase is located at the upper limit of the 2 to 4 $^{\circ}\text{C}$ temperature change projected by other studies in central Chile (DGF-CONAMA, 2007; IPCC, 2007).

Note that the eastern patch of the sclerophyllous forest has temperatures that are 1 or 2 $^{\circ}\text{C}$ lower than those of the western side. This is caused by the difference in altitude between the two forest patches.

Fig. 7 shows the projections of the SWB for the GCM INM-CM4-8, which is obtained from climatic projections (precipitation and temperature from Figs. 5 and 6). Note the predominance of negative amounts of

SWB, both with this GCM and the rest (see Fig. S10 to S13 of the Supporting Information Appendix S3). The reason why SWB is negative, is that it is estimated using PET, which is associated with potential evapotranspiration and not actual evapotranspiration. Although negative water balance does not have much physical sense, SWB can be used as a reference to the bioclimatic water stress that integrates both precipitation and temperature variables. Fig. 7 shows that quadrants 1, 2, 3 and 5 present amounts with similar trends, beginning the historical period with values between 0 and -200 mm approximately, which gradually decrease to SWB values of -500 mm by the end of century. On the other hand, Q4 and Q6 begin the historical period with positive SWB close to 100 and 500 mm, respectively, which drop to -500 and 0 mm by the end of the century.

Overall, Q6 stands out for presenting significantly higher amounts of precipitation and lower temperatures compared to the other quadrants

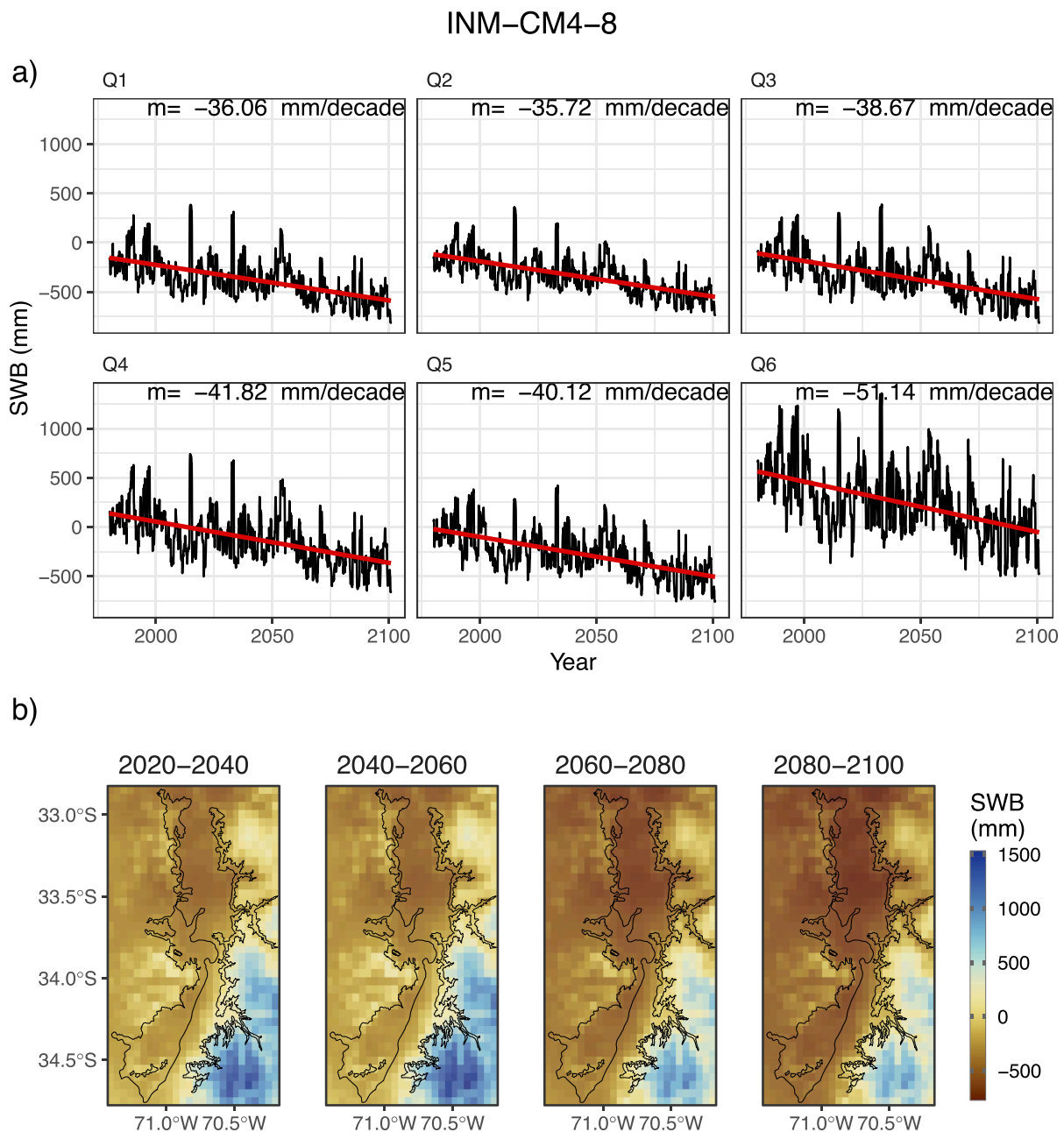


Fig. 7. Simplified water balance (SWB) of INM-CM4-8. A) Average time series for each quadrant, and its linear trend (red line). B) Spatial changes in SWB using 20-years windows between 2020 and 2100.

(Figs. 5 and 6). Also, the SWB adopts a behavior of space–time changes similar to that of precipitation (Fig. 7), hence precipitation is the dominant factor that has the greatest influence on the spatial distribution of SWB in the study area.

3.2. Critical threshold

The critical thresholds of SWB, used to detect the ToE of the forest, are presented in Fig. 8. These thresholds are obtained by using the 10th and 25th percentiles (i.e. P10% and P25%, respectively) of each location for the historical SWB (Fig. 3c). The values of P25% thresholds go from –600 to 1000 mm, while the P10% thresholds go from –650 to 750 mm.

Fig. 8 shows that the difference between both critical thresholds is in the range 50 to 100 mm. While the difference between the critical values between quadrant ranges from 300 to 500 mm, with Q1 being the driest,

and Q5 and Q6 the most humid.

3.3. Time of Emergence (ToE)

The most pessimistic climate projection (i.e. GCM CanESM5), shows that the forest will be affected by the period 2050–2060 in Q3, Q5 and the southern part of Q2 (Fig. 9). By that period the first threshold will be exceeded, probably showing serious damage and changes in terms of the forest structure, composition, dynamics and functions. By the 2060 s, the entire forest will have passed the P25% threshold in this climate projection, and also most will have exceeded P10% threshold, except for a few sites in Q1, Q4, and Q6. Finally, by the 2070 s the development of the forest is no longer feasible in its initial distribution area (i.e. the P10% threshold will have been exceeded).

The IPSL-CM6A-LR GCM presents similar results than CanESM5 after 2060 s. Quadrants 3 and 5 already show serious deterioration symptoms

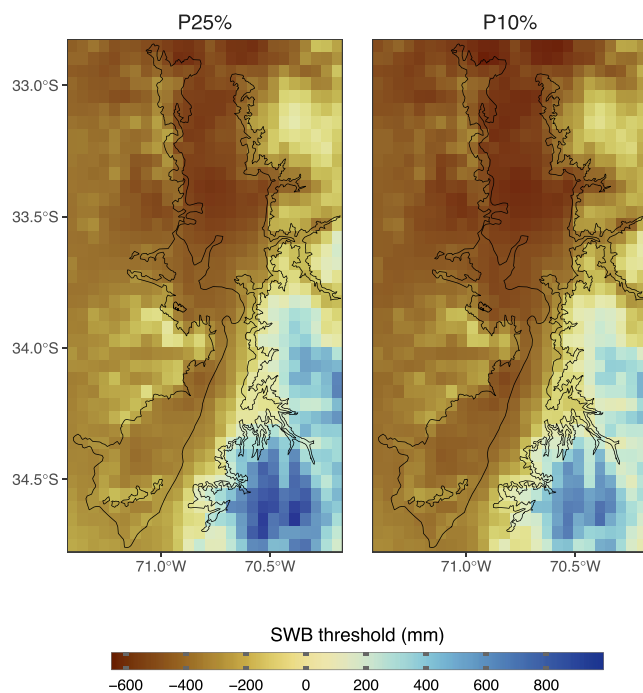


Fig. 8. Critical thresholds P25% and P10% of the historical SWB for the study area.

due to climate change, which rapidly spread to the entire forest after the years 2070–2080, but with a great difference in the magnitude of the impact compared to CanESM5. During 2070–2080, almost the entire forest will have exceeded the P10% threshold for IPSL-CM6A-LR, except for a few sites in the foothills of quadrants 5 and 6, which present a ToE for the 2080 s and 2090 s.

The results of the ToE of the GCM INM-CM4-8, indicate that between the years 2070 and 2080 the first critical thresholds (P25%) will be exceeded in a large part of quadrants 2, 3 and 5. This is around 20 years later than the ToE detected with the GCM CanESM5. During the period 2080–2090, the damage will have spread to the entire forest, and in that same decade, the forest will be almost completely displaced in all quadrants except for Q6. By the years 2090–2100, even in this intermediate climate change projection, the development of the forest will be present only in the southern-cordilleran sector and a fraction of Q4, showing a great change, with respect to its current distribution.

Under SSP2-4.5 scenario, the CESM-WACCM GCM projection is notoriously more favorable than the projections under SSP5-8.5. It is only at the end of the century that the forest will begin to be seriously affected, which under the pessimistic projections of SSP5-8.5 occurs around the years 2060–2070. The distribution of the forest according to this model will not be compromised by the year 2100. Note that the fact that there is no ToE of the SWB in this century, means that it may happen afterwards or it may not occur.

According to what is projected by the GCM NorESM2-LM, the most optimistic GCM evaluated here, the SWB₃₀ does not exceed the thresholds during this century. The dispersion of the ToE results shows the great uncertainty within climate change projections, where the most pessimistic projection shows an early ToE of the SWB₃₀ (i.e. during 2060–2070), while a more optimistic projection the ToE may not even

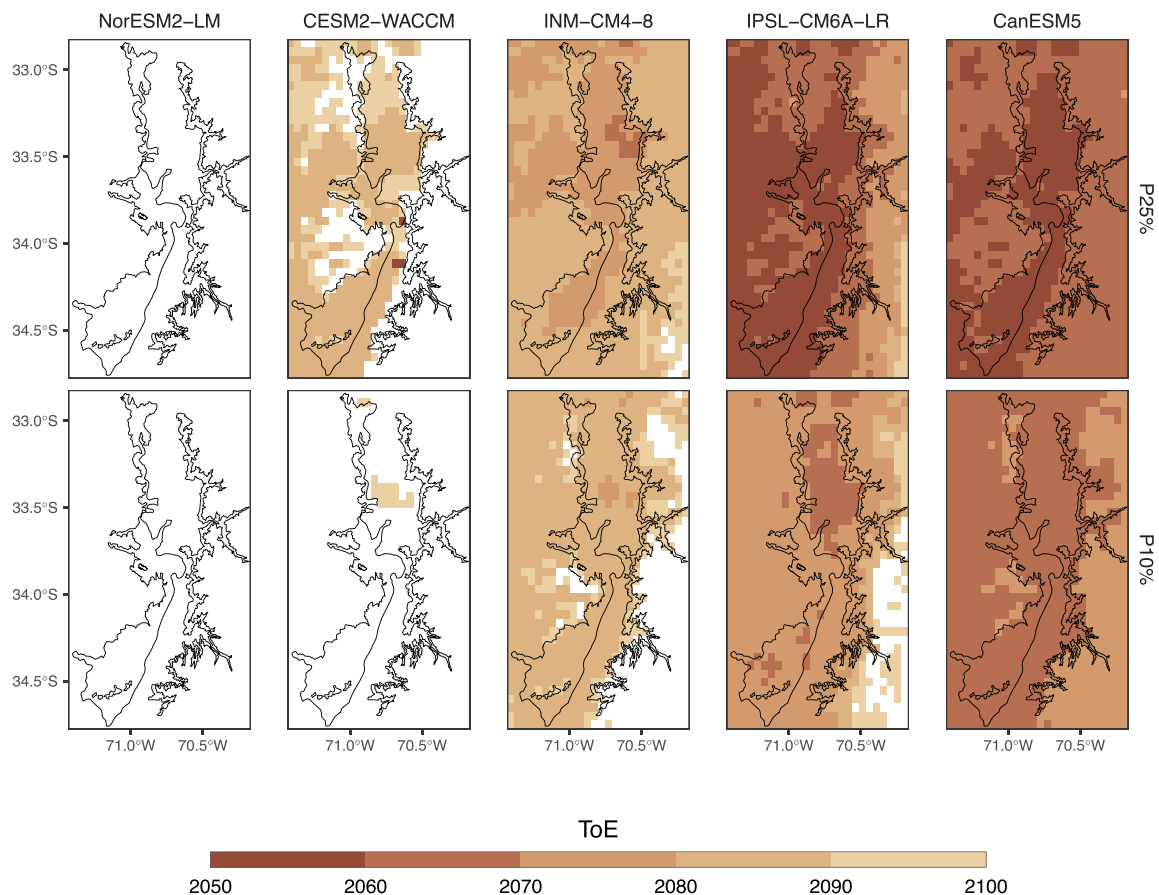


Fig. 9. SWB₃₀ Times of Emergence (ToE) for each GCM and critical threshold (P10% and P25%).

occur.

It should be noted that despite the fact that Q1 presents more restricted humidity conditions than other sites in the sclerophyllous forest, it projects a later ToE. This has the assumption that the forest is expressing itself differently, according to the mesoclimatic conditions in which it is found, and hence can have different responses to the same climatic condition. This is the reason why some sites in Q6 present an earlier ToE, than other sites in Q1. Although the SWB in Q6 might have bigger values, it is projecting a faster decrease in water availability, which should affect the forest in those sites (Fig. 7a and Fig. S9 to S12). We also added an alternative assumption in the [supplementary information](#) (Appendix S4), by evaluating the ToE of the forest by considering a uniform critical threshold throughout its extension. Under this alternative assumption, the most pessimistic projections show that the forest would begin to be seriously affected between 2020 and 2050 for quadrants 1, 2, 3 and 5. While in quadrants 4 and 6 the forest would be affected between 2070 and 2080. While for the most moderate projections, some sites located in Q1 have already exceeded the first threshold (25%) between 2010 and 2020, and they will exceed the development threshold (10%) during the 2030 s. For more information on the ToE of the forest under this alternative assumption, go to the [supplementary material](#) (Appendix S4).

4. Discussion

The precipitation projections we have obtained (Fig. 5) agree with previous studies, which use previous versions of the GCMs. An example, is a central Chile (32–35°S, 70.5–71.5°W) study that analyzed 30 GCMs from CMIP5 (Matskovsky et al., 2021), obtaining annual precipitation that started in 500 mm in 2005, which diminished to ~ 400 mm by the 2065, under RCP 8.5. Similarly, Chadwick et al. (2018) analyzed future climatic conditions for three Chilean basins using 45 GCM projections, and obtained that in the Maipo River basin precipitation is expected to decrease from 450 mm in 2005, to 350 mm at the end of the century, under the RCP 8.5.

In temperature, the results found (Fig. 6) are also similar to those presented by Matskovsky et al. (2021) and Chadwick et al. (2018). Matskovsky et al. (2021) study had an increase in temperature from 12 to 14.5 °C between the period 2005–2065, while the results of Chadwick et al. (2018), projects temperature increases of 3.8 °C from 2005 to the end of the century, both studies under RCP 8.5.

The results show that the forest sites most vulnerable to climate change are those located in quadrants 1, 2, 3 and 5 (Fig. 9). The changes in the forest would begin with Q2, specifically in the foothills of Santiago, and in the southernmost sites of Q5, almost immediately followed by Q3. These changes are identified using the critical threshold P25%, and occur between 2050 and 2080 under scenarios SSP5-8.5 (pessimistic) for GCMs CanESM5 and INM-CM4-8, respectively. On the other hand, under more moderate scenarios, the change occurs around the year 2090, or may not even happen during this century. The ToE of changes in the distribution (P10%) are expected to occur between the years 2060 and 2090 under the pessimistic scenarios (SSP5-8.5). While in the moderate scenario no distribution changes are projected (GCMs NorESM2-LM and CESM2-WACCM, under SSP2-4.5 in Fig. 9).

These results differ from those obtained under the methodological approach that uses a homogeneous threshold for the entire forest ([Supporting Information](#) Appendix S4), mainly in Q4 and Q6, and in the ToE of the first threshold (P25%). When using a single threshold for the entire forest, the threshold is more difficult to exceed in wetter sites such as Q4 and Q6. The single threshold for the entire forest in the detection of ToE using P25% is related to the symptoms of browning that are currently occurring in the forest, due to the megadrought. This leads us to an early ToE occurring even by 2020 in some sites of Q1, Q2 and Q3, contiguous to the city of Santiago (Fig. S14, [Supporting Information](#) Appendix S4), where almost all browning points have been identified (Fig. S1). However, the places where this occurs are only a few, since, in

the rest of Q1, Q2 and Q3, the ToE detected is similar regardless of which of the methodological approaches is used. The same happens with the ToE of the second threshold (P10%), where both project distribution changes between the years 2060 and 2090 under pessimistic scenarios (CanESM5 and IPSL-CM6A-LR, under SSP5-8.5) in Q1, Q2 and Q3.

The forest presents a degree of threat compared to the surrounding vegetation belts. Although this forest has some mechanisms to overcome droughts, the results show that climate change will present itself faster in the forest under study than in the surroundings. Only in the city of Santiago (city in the central valley, between the two patches of forest, see Fig. 1) there are sites where the thresholds are exceeded before those from the forest, the rest of the surrounding areas of the forest present changes in climate that will occur later (i.e., the central valley of Chile is between both patches of forest, the Cordillera de los Andes at the east and the Cordillera de la Costa at the west). These projections agree with the greater intensity that the megadrought has had in the central valley (Garreaud et al., 2017), which has so far caused greater browning in sclerophyllous forests towards the Andes Mountains, such as the sclerophyllous forest Mediterranean Andean forest of *Q. saponaria* and *L. caustica* than in coastal forests such as the Mediterranean Coastal Sclerophyllous Forest of *Cryptocarya alba* and *Peumus boldus*, and the Mediterranean Coastal Spiny Forest of *Acacia caven* and *Maytenus boaria* (Miranda et al., 2020b). These results and those of Miranda et al. (2020b) indicate that the vegetation belt under study is more vulnerable than neighboring belts, with higher browning and earlier ToE. However, it is not clear whether the adaptive characteristics of the forest against droughts represent a competitive advantage over formations belonging to the deciduous forest, and other sclerophyllous formations. A study by Plischoff et al. (2012), projected under scenarios B2 and A2 of the third IPCC report, that by the year 2080 the sclerophyllous forest would move south, populating over 50% of the territory today occupied by the deciduous forest. On the other hand, Bambach et al. (2013), points out that the Andean Mediterranean Sclerophyllous Forest of *Q. saponaria* and *L. caustica* would have a greater potential of climate change adaptation than certain coastal sclerophyllous vegetation belts. Hence, the fact that climate change impact manifest earlier on a forest, is not necessarily related to a greater damage in the future. In any case, the impacts the forest will have in the future depend on the capacity of the forest to migrate (Bambach et al., 2013), so it is very important to avoid forest and scrub fragmentation (Hernández et al., 2015).

Although, the purpose of this study is not to model the future distribution of the forest, and there are more adequate methodologies for this; it is possible to estimate possible sites to which it could migrate. Also, this methodology might help determine sites that need conservation, restoration or management efforts, as well as, where the migration process should be supported. Considering that the Andean Mediterranean Sclerophyllous Forest of *Q. saponaria* and *L. caustica* has a greater adaptive capacity to climate change, unlike coastal sclerophyllous forests (Bambach et al., 2013), it could be expected that the western patch of the forest presents movements towards that sector. Spatial movement to a new location could occur towards what is now the Mediterranean Coastal Sclerophyllous Forest of *Cryptocarya alba* and *Peumus boldus*, and the Mediterranean Coastal Deciduous Forest of *Nothofagus macrocarpa* and *Ribes punctatum* (see Fig. S15 of [Supporting Information](#) Appendix S5), given that in the sites where they are located, the effects of climate change on water resources generally appear later. On the other hand, the eastern patch will most likely respond with a longitudinal movement pattern, like many other species or ecosystems in warm areas, modifying its distribution to areas of higher altitude (Pech et al., 2017). This means that it most likely will come to occupy part of the territory where the Andean Mediterranean Sclerophyllous Forest of *Kageneckia angustifolia* and *Guindilia trinervis* is located today (see Fig. S13 of the [supplementary material](#)). However, it must be considered that the movement of the forest will be determined mainly by landscape metrics (Dener et al., 2021) and the optimal climatic conditions for the establishment of seedlings. Therefore, there is no certainty that the forest has the

possibility of migrating, in the future, to other sites with more favorable conditions.

The displacement of the forest caused by the diminishing water resources will occur later for quadrants 4 and 6, because they are the areas with the highest water availability. Muñoz-Sáez et al. (2021) had identified these places as one of the two climatic refuges for the sclerophyllous forest, the other is located in the coastal mountain range at latitude 33°S. As mentioned by Hannah et al. (2007), for a protected area to be effective in the long term, climate change studies must be carried out as soon as possible and included in the planning. For this reason, we recommend carrying out, conservation actions that protect these ecosystems from human disturbances, such as the widening of the R.N. River of the Cypresses, R.N. Clarillo River (Fig. 1), or the creation of new protected wild areas in those quadrants. On the other hand, in the rest of the quadrants where future water conditions are more unfavorable, several hydrological-forestry management techniques should be incorporated, such as the combination of soil and water conservation works and rainwater harvesting. These would be an option to increase water availability by reducing runoff and favoring infiltration, which can be supported together with the incorporation of arboreal and shrubby plant species as the quality of the remaining soil allows it (Labra et al., 2018).

The methodology proposed here makes it possible to quickly, cost-efficiently and easily generate information capable of supporting decision-making in natural forests. Being a contribution in bridging the gap between scientists and decision makers (Grassi et al., 2017). The methodology and results are not intended to replace the studies of distribution changes, given that the objective is not to simulate future forest distributions, but rather to complement them with a simple implementation that will allow to understand the impacts of climate change over forest and ecosystems. The results have several sources of uncertainty, being one of them the thresholds adopted in the methodology. Unfortunately, the tolerance thresholds and the various responses of the forest to climatic variability and change are unknown, which occurs not only to sclerophyllous forests, but also to many other biological systems (Santibáñez et al., 2013). Therefore, in subsequent studies the results here obtained should be complemented with species distribution modeling studies, and above all, carry out studies to improve our understanding of the response the forest and its species has to changes in climate. Forest monitoring is of particular relevance in the context of global changes, since it is impossible to understand the relationship forest-climate without it. Increasing this knowledge will allow to improve climate change mitigation and adaption strategies, and the conservation and restoration of biodiversity (Holmgren et al., 2007; MacDicken, 2015); which is needed to anticipate changes proactively (Root et al., 2003).

When estimating the ToE on a forest, there are collateral effects that climate change could have, caused by the interactions of other components of the system that could cause damage to the forest. The most notorious example are wildfires, which have caused deterioration over Chilean vegetation. Although the causes are mainly human activities, climatic conditions have an impact on the spread and maintenance of fires (Pica-Téllez et al., 2020). Temperature increase and the rainfall reduction in central Chile, due to the megadrought (Boisier et al., 2016), has considerably increased the availability of dry biomass. Generating favorable conditions for rapid ignition and the spread of wildfire, such as the one seen in the mega forest fires that occurred in 2017 (Castillo et al., 2020). In the last decade (2008–2018), 16 of the 22 mega fires registered since 1985 have occurred, which has caused the annual burned area of native forest to almost double (González et al., 2020). Unfortunately, the sclerophyllous forest is not a fire-dependent forest, reason why it has significantly suffered due to recent fires, presenting an important deterioration. Therefore, it is expected that the number of fires, their duration, and the duration of the season, will remain at this higher value of the last decade or it might even increase in the future (González et al., 2018). If this happens, the deterioration of the forest and its migration

might take place even before than the detected ToE of this study. Another possible indirect effect of climate change over forests, are land use changes, caused by the detriment in crop productivity (Thapa, 2022). It is also possible that alterations in the plant-herbivore interaction, might affect the susceptibility of the forest to pathogens and their growth, reproduction and infection (Sturrock et al., 2011). These climate change collateral effects, can have important repercussions over on the Mediterranean forest of central Chile. For simplicity, climate change collateral effects have not been addressed in this study, but they must be taken into account to understand the large number of factors linking climate-forest interactions that influence ecosystems. Therefore, it is possible that the ToE of the sclerophyllous forest might occur earlier than expected in the results.

5. Conclusion

This study presents a methodology to estimate when and where changes will arise in the distribution of a Mediterranean forest due to climate change, which is applied to the Andean Mediterranean sclerophyllous forest of *Quillaja saponaria* and *Lithraea caustica*. This is done from the cross between five climate projections of a simplified water balance (BHS) and critical thresholds of the forest under study. When critical thresholds are exceeded, it is considered as the ToE of the forest. Although the ToE results vary between GCMs and certain methodological aspects, they show changes in the distribution between the years 2060 and 2090 for most of the forest, under the SSP5-8.5 scenario. Under the SSP2-4.5 scenario, no significant changes in its current distribution are projected by the end of the century. The results indicate that the first sites to be affected and displaced would be those located in the coastal mountain range next to the counties of Til Til and Lampa, and the foothills of Santiago, followed by localities located in quadrants 3 and 5, such as the Talagante, Isla de Maipo, Paine, Doñihue, and its surroundings.

The methodology proved to be adequate to develop quick, simple and accessible overall analysis, ideal for supporting decision-making in the areas of ecosystems conservation, protection, restoration and management. One can divide the sclerophyllous forest in two sectors, when considering its future water availability. Given these differences, decision-making requires different considerations for each sector. For example, sectors Q6 and part of Q4 require conservation and protection projects, since it more favorable water conditions will allow the forest to remain longer. While in the rest of the forest, where the water conditions are more unfavorable, it is advisable to incorporate hydrological-forestry management techniques, through the combination of soil and water conservation works, and rainwater harvesting, before any reforestation project is developed. This more favorable conditions are required to avoid the deterioration of the forest or the failure of restoration plans implemented in the area.

We would also like to highlight that the proposed methodology to detect the ToE in Mediterranean forests, can be extended to other ecosystems, by adapting the climatic variable or index to that specific location. The climatic variable or index of interest for other forests need to be something that limits the forest or ecosystem development and growth, for example: the simplified water balances, the index of aridity, heat waves, extreme temperatures, meteorological indices of fires, among others. Greater knowledge about ecosystems and their responses to climate change and variability allows us to reduce methodological uncertainties in this type of study. In particular, the methodology here proposed can be applied to more specific ecosystems, such as vegetation belts, whose limited distribution does not necessarily account for the climatic tolerance of the different species that comprise it. Additionally, future studies should compare and complement the results here obtained with those of species distribution modeling, given that the main objective of the estimation of ToE over forest is understanding climate change impacts over a forest and not the distribution simulation.

CRediT authorship contribution statement

David Martínez: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Cristián Chadwick:** Conceptualization, Methodology, Software, Supervision, Writing – original draft, Project administration, Writing – review & editing. **Andrés Plaza-Aguilar:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2023.121169>.

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